

ELL-HALER, Netherlands

WMO: 063770

Lat: 51.1967N

Lon: 5.7625E

Elev: 30

StdP: 100.97

Time Zone: 1.00 (EUC)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-6.5	-4.4	-10.1	1.6	-5.4	-7.3	2.0	-3.3	12.8	8.6	11.5	9.3	2.6	50	0.536

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.6	29.9	20.6	27.7	19.6	25.8	18.6	21.6	27.8	20.5	26.1	19.6	24.4	3.1	100

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.5	14.3	24.0	18.6	13.5	22.7	17.7	12.7	21.9	62.9	27.7	59.2	26.3	56.0	24.4	26.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
9.9	8.6	7.6	-9.0	34.3	4.1	2.1	-12.0	35.8	-14.3	37.0	-16.6	38.2	-19.6	39.7		
			DB	WB	DB	WB	DB	WB	DB	WB	DB	WB	DB	WB		
			-9.5	24.2	3.9	1.0	-12.3	24.9	-14.6	25.5	-16.8	26.1	-19.6	26.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	10.8	3.3	3.9	6.6	10.2	13.8	17.1	18.6	17.8	14.8	11.4	7.0	4.2
	DBStd	6.48	4.33	4.04	3.58	3.64	3.51	3.10	3.11	2.68	2.84	3.55	3.45	4.12
	HDD10.0	857	211	172	114	43	7	0	0	0	1	27	101	183
	HDD18.3	2868	467	403	364	246	146	60	36	42	112	216	338	439
	CDD10.0	1134	2	2	9	48	125	212	266	241	145	70	12	3
	CDD18.3	103	0	0	0	0	6	22	43	25	6	1	0	0
	CDH23.3	1089	0	0	0	18	75	230	431	260	71	4	0	0
	CDH26.7	301	0	0	0	1	13	58	134	82	13	0	0	0
Wind	WSAvg	3.5	4.4	4.2	4.0	3.4	3.4	3.1	3.0	2.9	2.8	3.3	3.7	4.2
Precipitation	PrecAvg	772	65	61	49	43	58	72	76	82	60	63	68	76
	PrecMax	978	131	148	96	97	103	198	168	180	154	156	114	152
	PrecMin	611	6	11	2	0	18	12	30	16	12	32	7	29
	PrecStd	93	31	31	26	22	25	39	36	47	37	25	26	31
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.2	15.0	19.7	26.3	28.3	31.3	33.5	32.5	28.6	23.4	17.5	14.0
		MCWB	11.0	10.1	12.3	15.9	18.9	21.5	21.7	21.8	19.7	17.7	14.4	11.8
	2%	DB	11.9	12.3	16.9	22.7	25.3	28.3	30.1	28.9	25.6	20.2	15.1	12.5
		MCWB	10.4	9.7	11.2	14.3	16.6	19.7	20.8	20.7	18.8	16.3	12.9	10.8
	5%	DB	10.4	11.0	14.4	20.0	23.2	26.0	27.7	26.1	23.0	18.1	13.6	11.2
		MCWB	9.0	8.9	10.4	13.2	15.7	18.6	19.8	19.2	17.7	15.1	11.9	9.8
	10%	DB	9.1	9.7	12.4	17.4	21.0	23.8	25.6	23.8	20.7	16.5	12.4	9.8
		MCWB	7.8	8.1	9.3	12.0	14.9	17.7	19.0	18.1	16.2	14.1	10.9	8.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.7	11.2	13.2	16.4	20.1	23.1	23.0	23.1	20.9	18.1	14.6	12.3
		MCDB	12.5	13.1	17.6	23.6	26.6	29.7	30.7	30.0	26.4	22.3	16.6	13.5
	2%	WB	10.4	10.2	11.9	15.0	18.1	20.7	21.6	21.2	19.4	16.5	13.2	11.1
		MCDB	11.6	11.7	15.3	21.4	22.9	26.1	28.1	27.1	24.3	19.5	14.8	12.2
	5%	WB	9.1	9.3	10.8	13.7	16.7	19.5	20.5	20.1	18.1	15.5	12.1	9.9
		MCDB	10.3	10.9	13.7	18.9	21.2	24.3	26.2	25.2	22.1	17.8	13.5	11.1
	10%	WB	7.9	8.0	9.7	12.5	15.5	18.4	19.5	18.9	16.9	14.5	10.9	8.7
		MCDB	9.1	9.6	12.3	16.9	19.7	22.7	24.4	22.6	20.0	16.3	12.2	9.8
Mean Daily Temperature Range	5% DB	MDBR	4.9	6.1	8.4	10.9	10.2	10.6	10.6	10.4	10.3	7.9	5.9	4.8
		MCDBR	5.3	7.9	12.5	15.6	14.2	14.5	14.9	15.0	13.9	9.8	6.9	5.5
		MCWBR	4.6	5.7	7.7	8.2	6.7	7.0	6.6	7.3	7.7	6.3	5.2	4.5
	5% WB	MCDBR	5.4	7.1	10.9	13.8	12.0	12.5	13.1	13.1	12.6	9.4	6.4	5.2
		MCWBR	5.0	5.4	7.1	7.6	6.4	6.7	6.3	6.9	7.4	6.2	5.2	4.5
Clear-Sky Solar Irradiance	taub	0.329	0.352	0.384	0.418	0.424	0.423	0.430	0.421	0.392	0.367	0.342	0.325	
	taud	2.386	2.347	2.264	2.198	2.224	2.243	2.226	2.275	2.345	2.402	2.389	2.357	
	Ebn at Noon	684	765	805	819	831	835	821	808	790	739	657	619	
	Edh at Noon	63	84	109	130	133	132	132	120	100	79	62	56	
All-Sky Solar Radiation	RadAvg	0.75	1.43	2.58	4.13	4.95	5.37	5.06	4.32	3.20	1.88	0.89	0.58	
	RadStd	0.10	0.28	0.37	0.46	0.44	0.46	0.61	0.41	0.33	0.16	0.10	0.08	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (88 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page